

COVALENTX TECHNOLOGIES

Web Development Internship Program

**INTERNSHIP REPORT
OF
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Interactive Website Development

Five Hands-On Web Application Tasks

Technologies: HTML5, CSS3, JavaScript

Prepared For: CovalentX Technologies

Date: June 2026

Introduction

This report documents five progressive web development tasks completed as part of the Interactive Website Development program at CovalentX Technologies. Each task builds practical, real-world skills across front-end fundamentals, interactivity, API integration, and personal branding, using HTML, CSS, and JavaScript as the core toolkit.

The tasks move from foundational layout and interactivity work toward more advanced concepts such as asynchronous data fetching and full personal portfolio design, reflecting a structured, skill-building curriculum.



**FRONT-END
FOUNDATIONS****Task 01 | Responsive Landing Page**

A fully responsive hero landing page designed for a SaaS product, built to look polished and function smoothly across desktop and mobile devices. The focus was on a strong first impression through bold typography and a navigation bar that adapts as the user scrolls.

HTML5	CSS3	JavaScript
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Key Features

- Sticky navigation bar that changes background colour on scroll
- Mobile-first responsive layout using Flexbox/Grid and media queries
- Impactful hero typography and clear visual hierarchy
- Smooth, lightweight interactions powered by vanilla JavaScript

Outcome

Delivered a production-style landing page that adapts cleanly across screen sizes and demonstrates strong fundamentals in responsive design and scroll-based UI behaviour.

INTERACTIVE LOGIC**Task 02 | Tic-Tac-Toe Web Application**

An interactive Tic-Tac-Toe game built entirely with HTML, CSS, and JavaScript, allowing two players to compete by tracking game state and detecting winning combinations in real time.

HTML5	CSS3	JavaScript
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Key Features

- Click-based game board with real-time state tracking
- Win, loss, and draw detection logic across all winning combinations
- Clean, responsive grid layout for the game board
- Reset functionality to start a new round instantly

Outcome

Produced a fully playable, bug-free game that reinforces core JavaScript logic, event handling, and state management concepts.

NAVIGATION & UX**Task 03 | Stopwatch Web Application**

An interactive, fixed-position navigation menu paired with a functional stopwatch interface, where the menu dynamically changes style on scroll and on hover to improve usability across all pages.

HTML5	CSS3	JavaScript
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Key Features

- Fixed navigation menu visible at all scroll positions
- Dynamic color and font changes on scroll and hover events
- Start, pause, and reset stopwatch controls with accurate timing logic
- Consistent, accessible styling across menu states

Outcome

Built an always-visible, responsive navigation experience combined with precise time-tracking functionality, strengthening DOM manipulation and event-driven styling skills.

API INTEGRATION

Task 04 | Weather API Portal

A functional weather web app that retrieves real-time data based on user input or device geolocation, then presents it through a clean, easy-to-read visual layout.

HTML5	CSS3	JavaScript	Weather API	Geolocation API
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Key Features

- Real-time weather lookup by city search or browser geolocation
- Asynchronous API calls using Promises and async/await
- Graceful handling of loading states and failed requests
- Clean data visualization of temperature, conditions, and forecasts

Outcome

Delivered a reliable, user-friendly weather portal that demonstrates solid command of asynchronous JavaScript and third-party API integration.

PERSONAL BRANDING

Task 05 | Personal Portfolio Website

A personal portfolio website showcasing skills, projects, and accomplishments as a web developer, designed with an attractive, visitor-friendly layout from first impression to detailed background.

HTML5	CSS3	JavaScript
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Key Features

- Engaging home page with headline, professional photo, and skills summary
- Detailed “About Me” section covering background, education, and experience
- Project showcase highlighting key work and accomplishments
- Fully responsive, visually cohesive design across all sections

Outcome

Completed a polished, professional portfolio site that effectively presents skills and projects, ready to support real-world job and freelance opportunities.

Conclusion

The Interactive Website Development program at CovalentX Technologies provided a structured, hands-on journey through the core building blocks of modern front-end web development. Across five progressively challenging tasks, the work moved from foundational layout and styling concepts toward more advanced, real-world skills involving interactivity, live data, and personal branding — giving a well-rounded view of what it takes to build functional, user-ready websites and web applications.

The first task, the **Responsive Landing Page**, established a strong foundation in responsive design principles. Building a SaaS-style hero page with a scroll-reactive sticky navigation bar required careful use of CSS media queries, Flexbox/Grid layouts, and JavaScript scroll event listeners, while also reinforcing the importance of typography and visual hierarchy in creating a strong first impression.

The second task, the **Tic-Tac-Toe Web Application**, shifted focus toward interactivity and logic. Implementing click handling, game-state tracking, and win/draw detection across all possible combinations strengthened core JavaScript skills — particularly around DOM manipulation, conditional logic, and managing state without a framework.

The third task, the **Stopwatch Web Application with Dynamic Navigation**, combined two skills at once: building a fixed, scroll- and hover-responsive navigation menu, and implementing accurate start/pause/reset timing logic. This task emphasized real-time DOM updates and event-driven styling, both essential for building polished, professional-feeling interfaces.

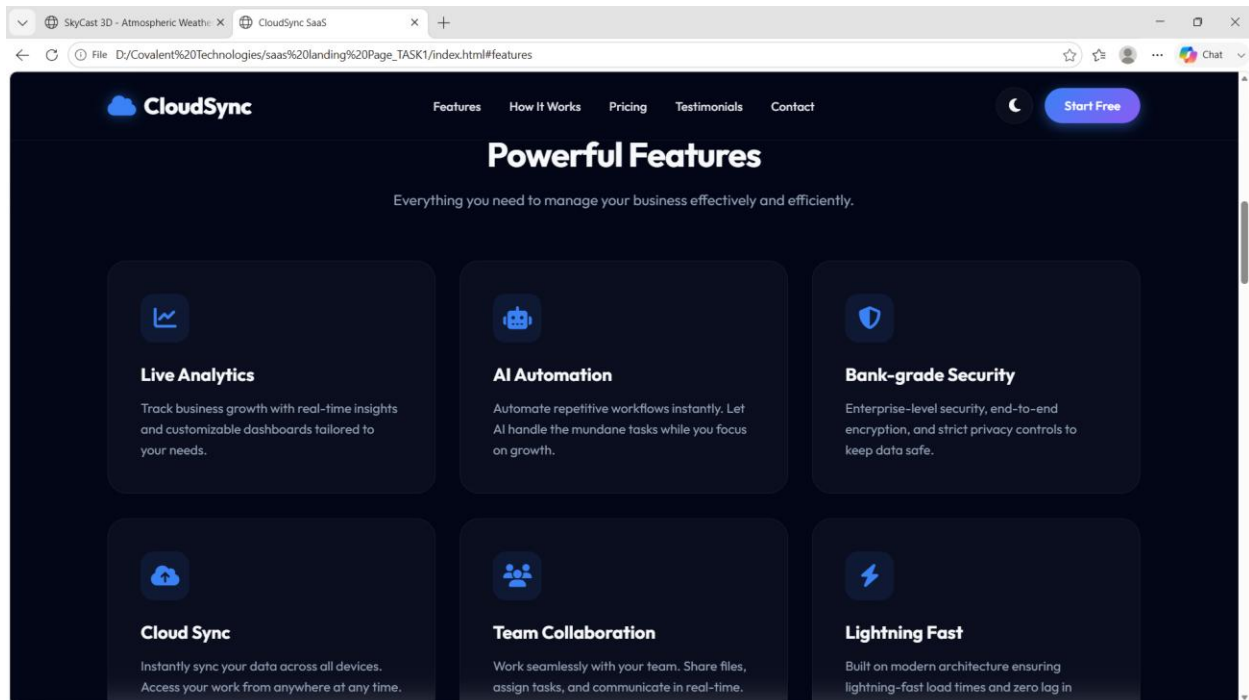
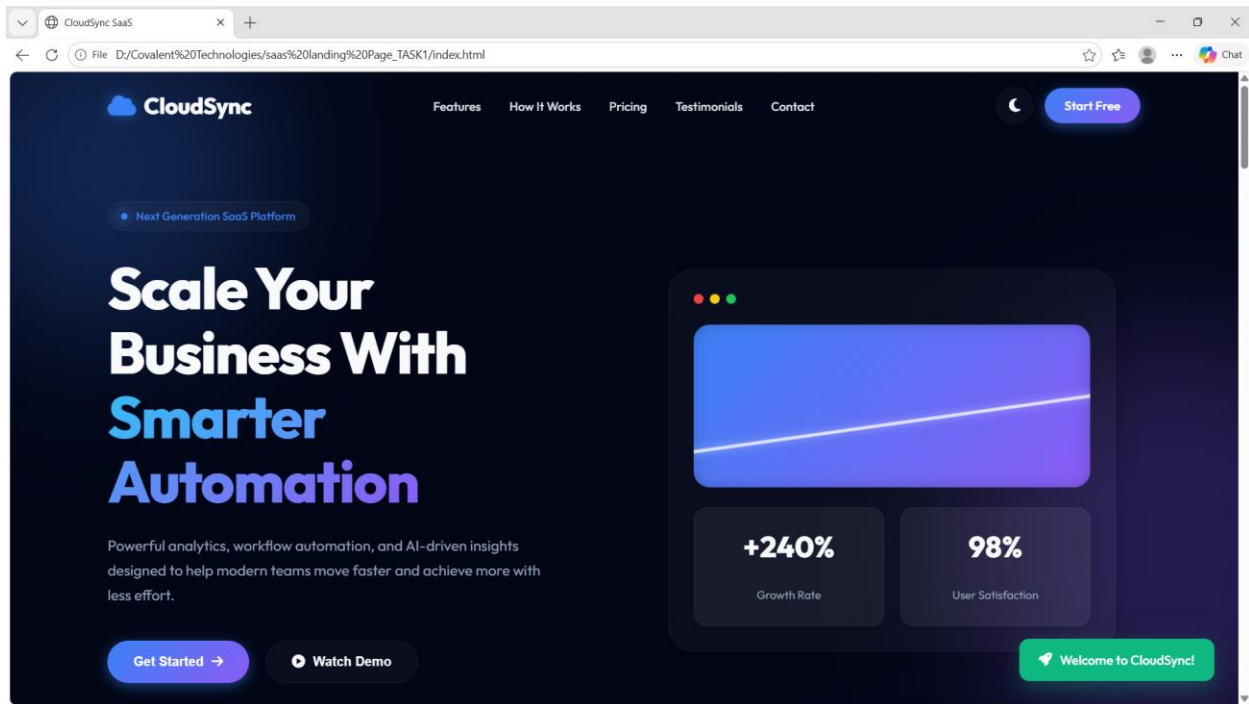
The fourth task, the **Weather API Portal**, introduced asynchronous programming and external data integration. Fetching real-time weather data based on user input or geolocation required working with Promises and `async/await`, handling loading and error states gracefully, and presenting the results through clean, readable data visualization — a meaningful step up in technical complexity compared to the earlier tasks.

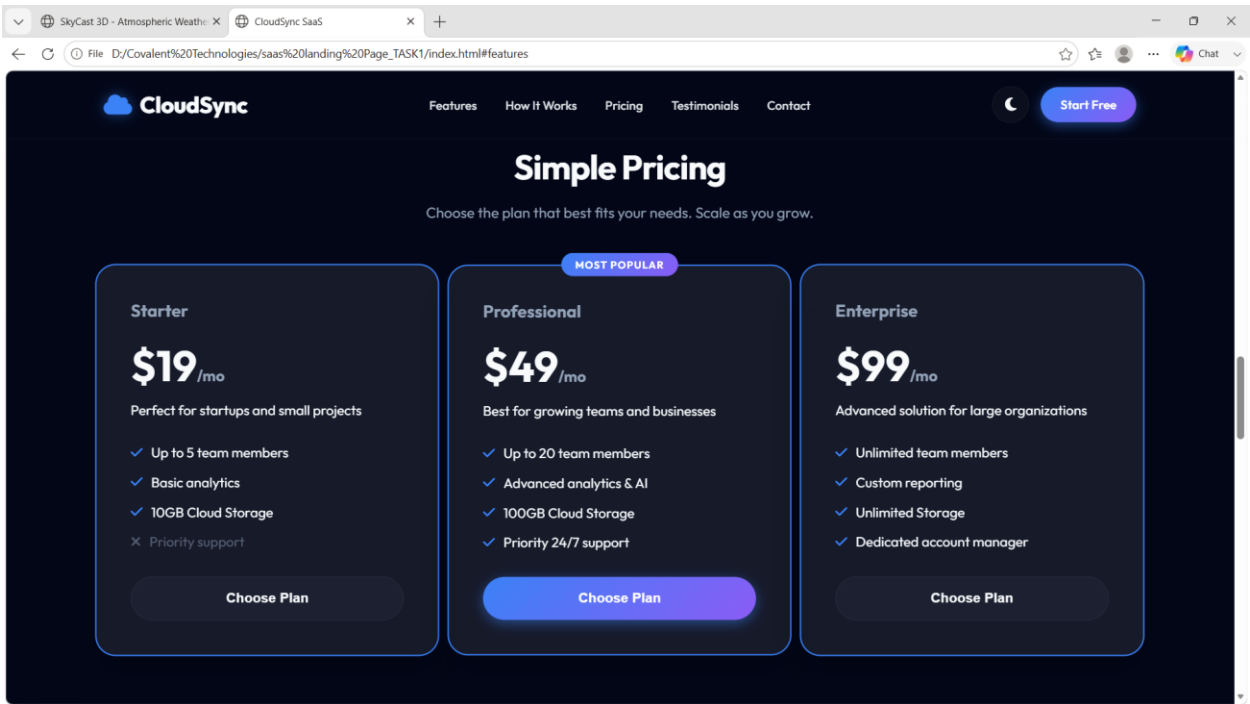
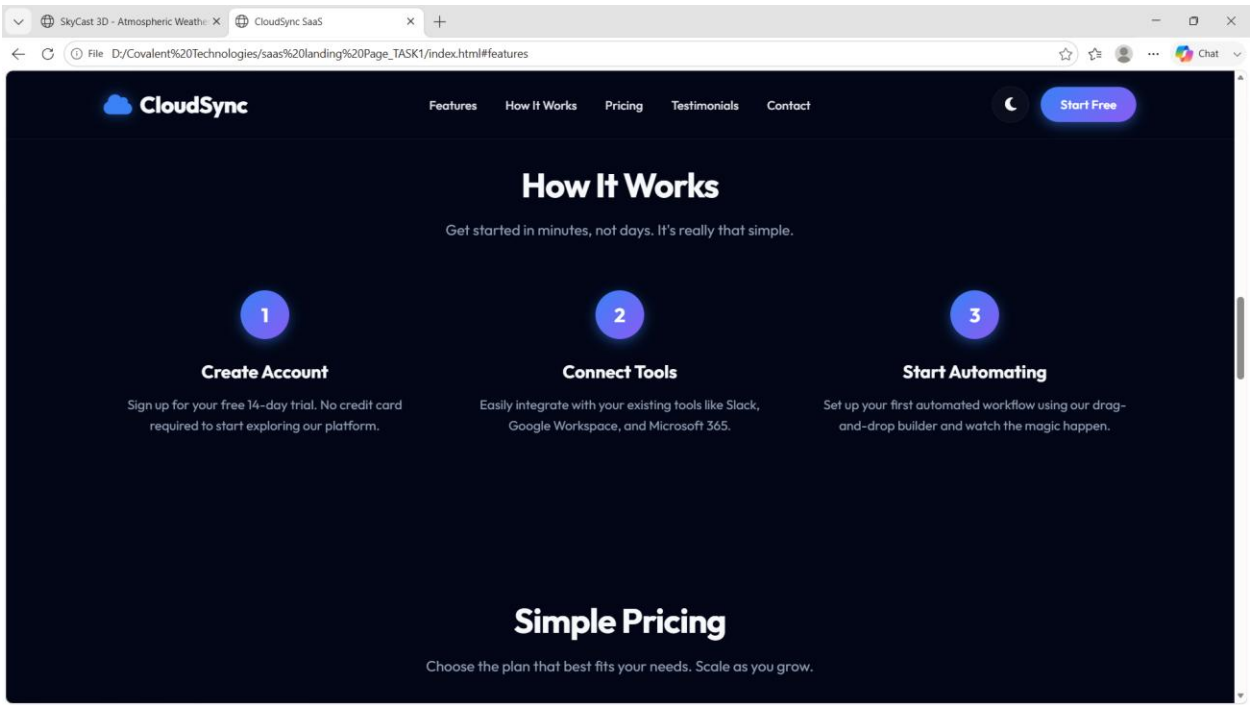
The fifth and final task, the **Personal Portfolio Website**, brought everything together into a cohesive, professional deliverable. It required not just technical execution but also thoughtful design and content decisions — crafting a compelling headline, structuring an "About Me" section with education and experience, and presenting projects and skills in a visually appealing, responsive layout.

Taken as a whole, these five tasks reflect a deliberate progression: from layout and responsiveness, to interactivity and logic, to UX-focused navigation, to live API-driven functionality, and finally to a polished personal showcase. Each project was built using only HTML, CSS, and JavaScript/TypeScript, demonstrating that strong fundamentals — without relying on frameworks — are sufficient to produce functional, attractive, and genuinely useful web applications. This experience has built a solid, demonstrable skill set in front-end web development, one that's directly applicable to real-world projects, freelance work, and future internship or job opportunities.

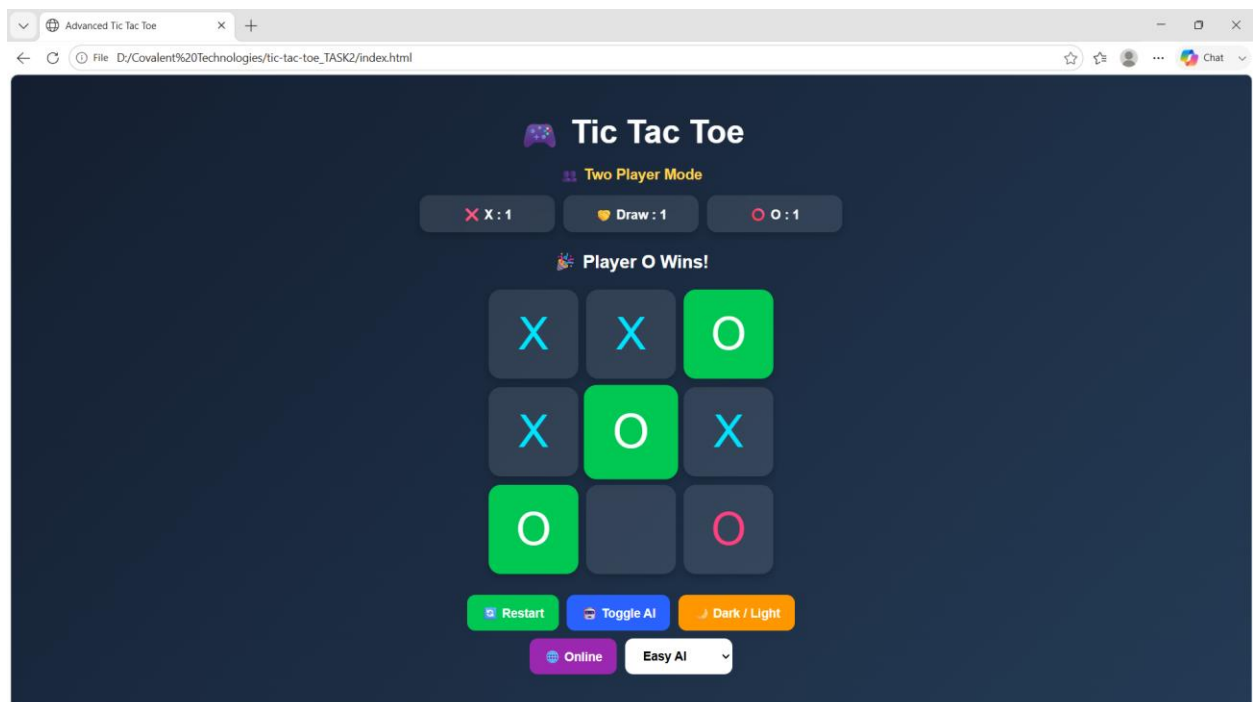
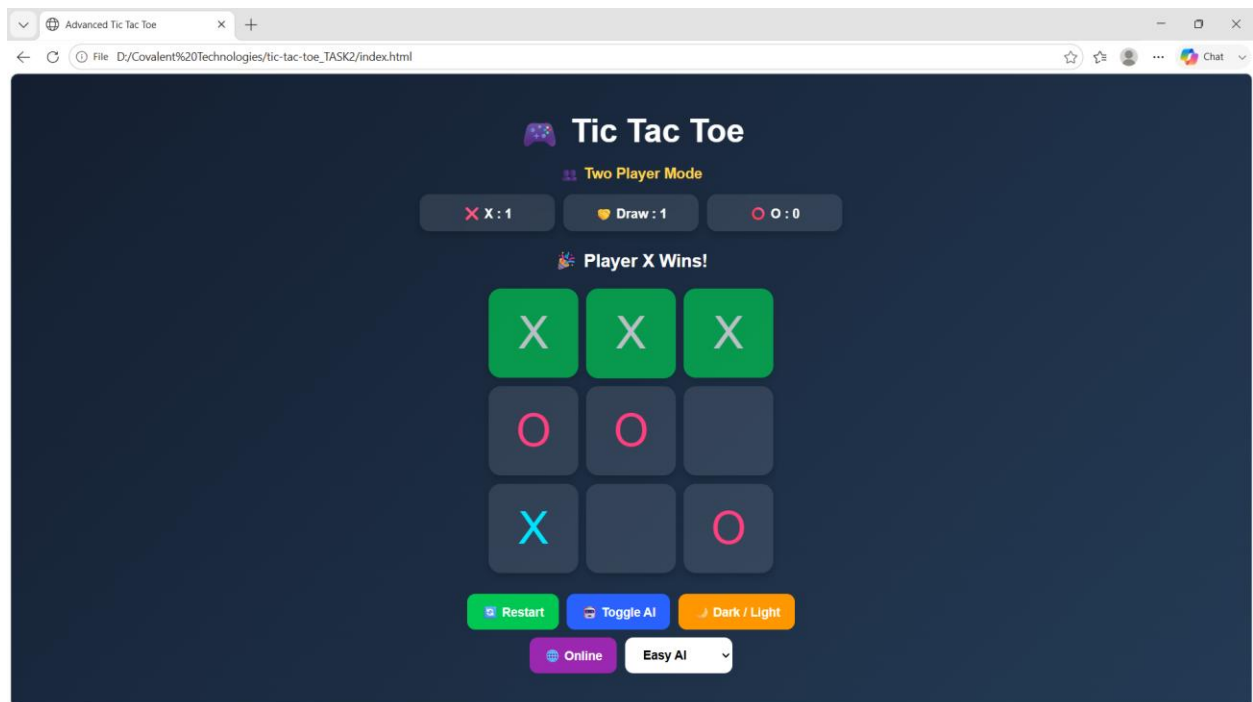
OUTPUT OF ALL TASKS:-

TASK 1:-

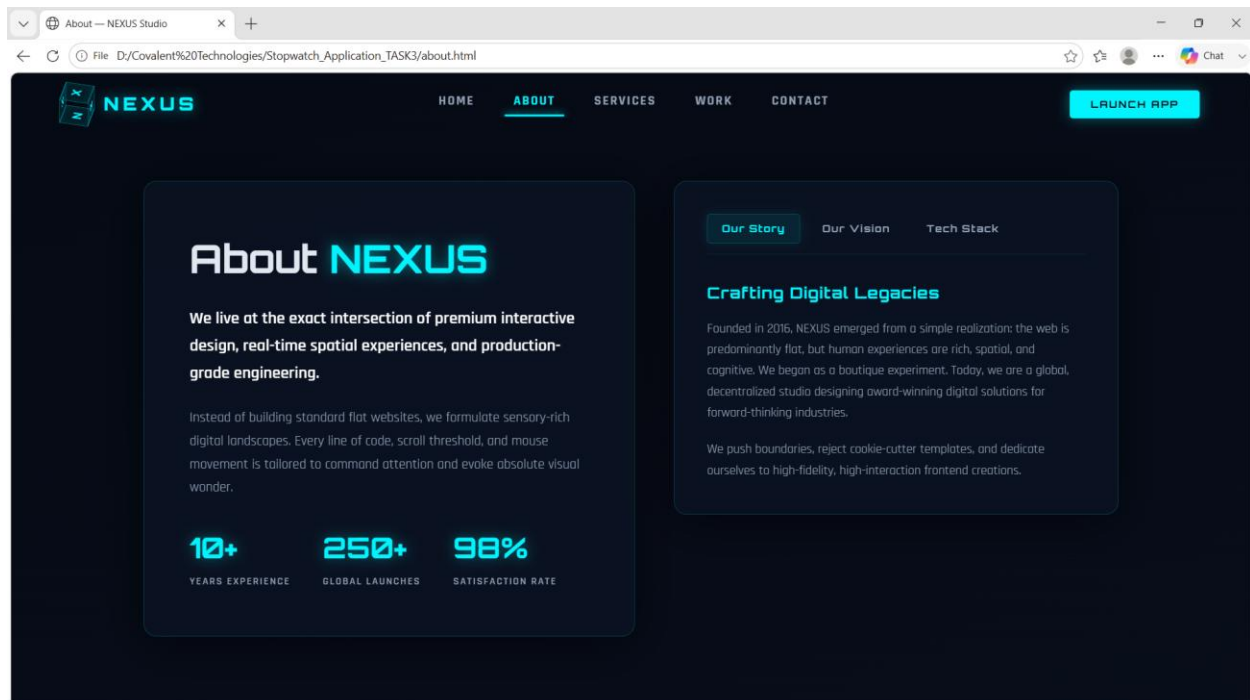
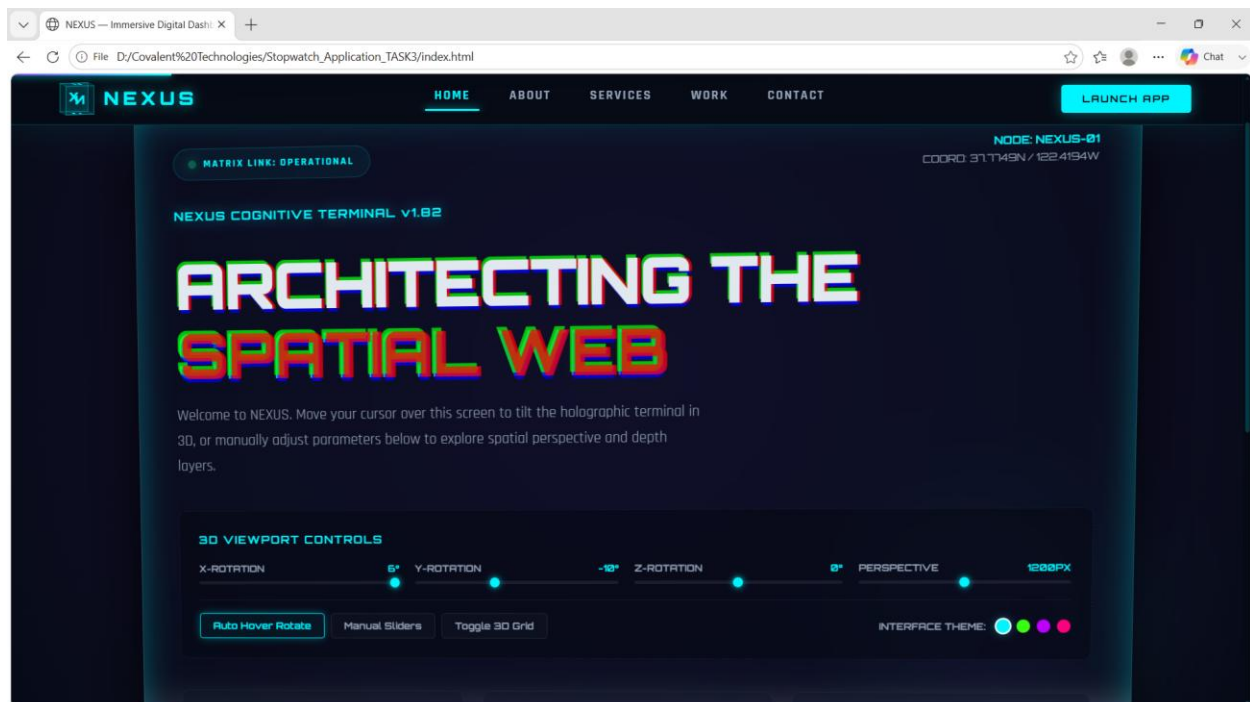


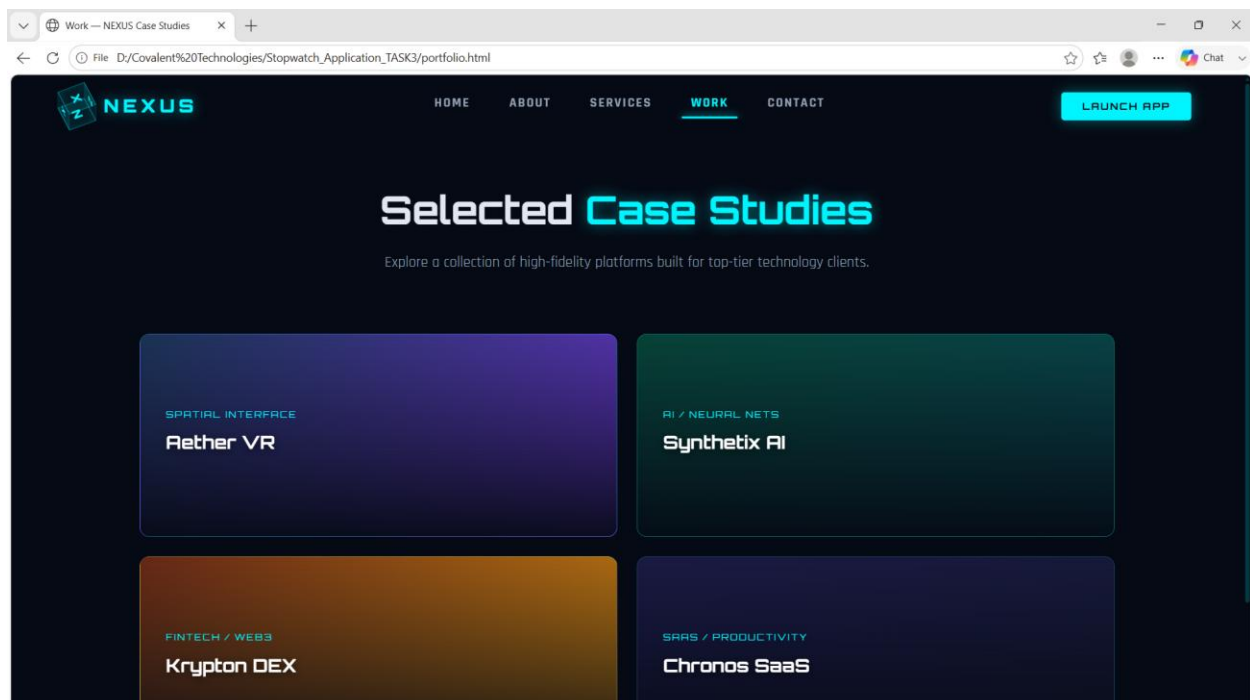
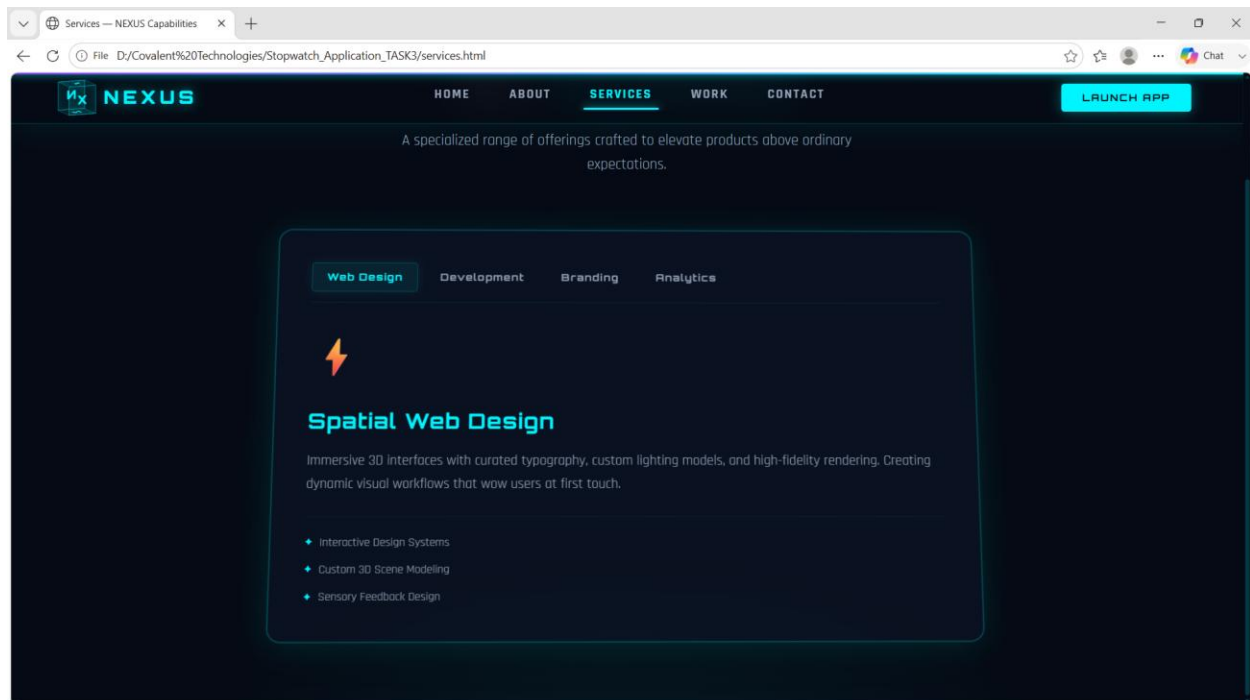


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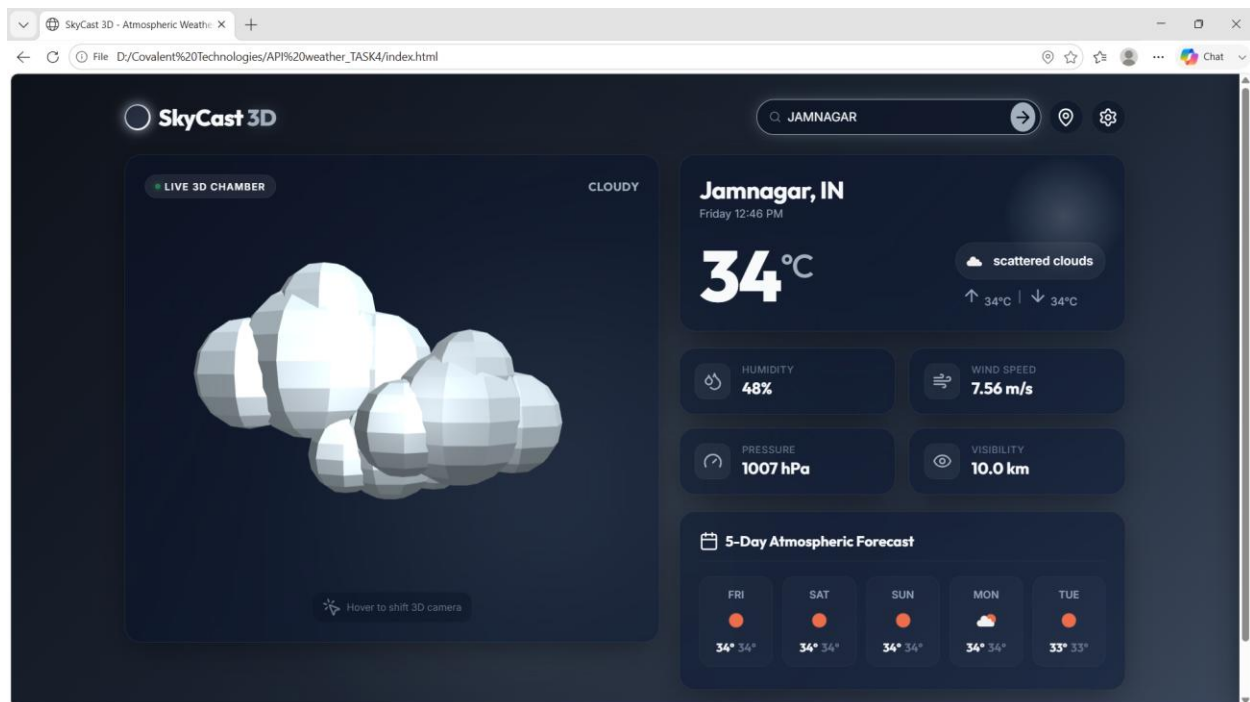


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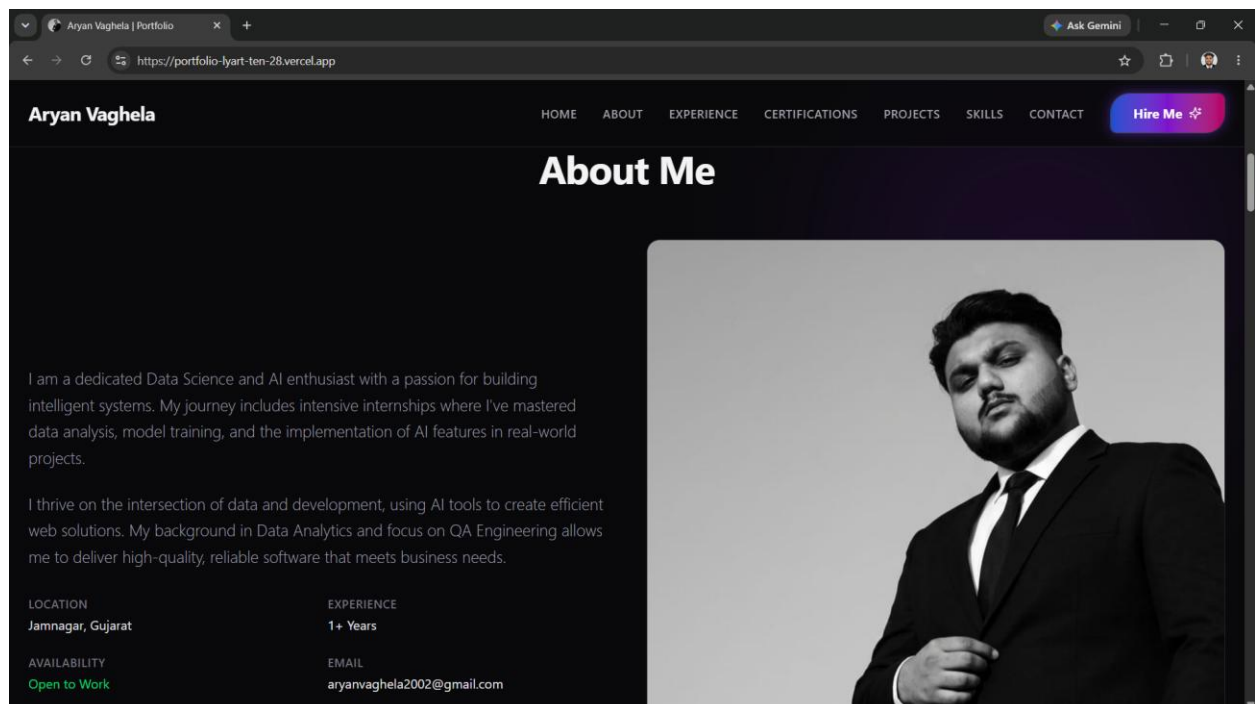
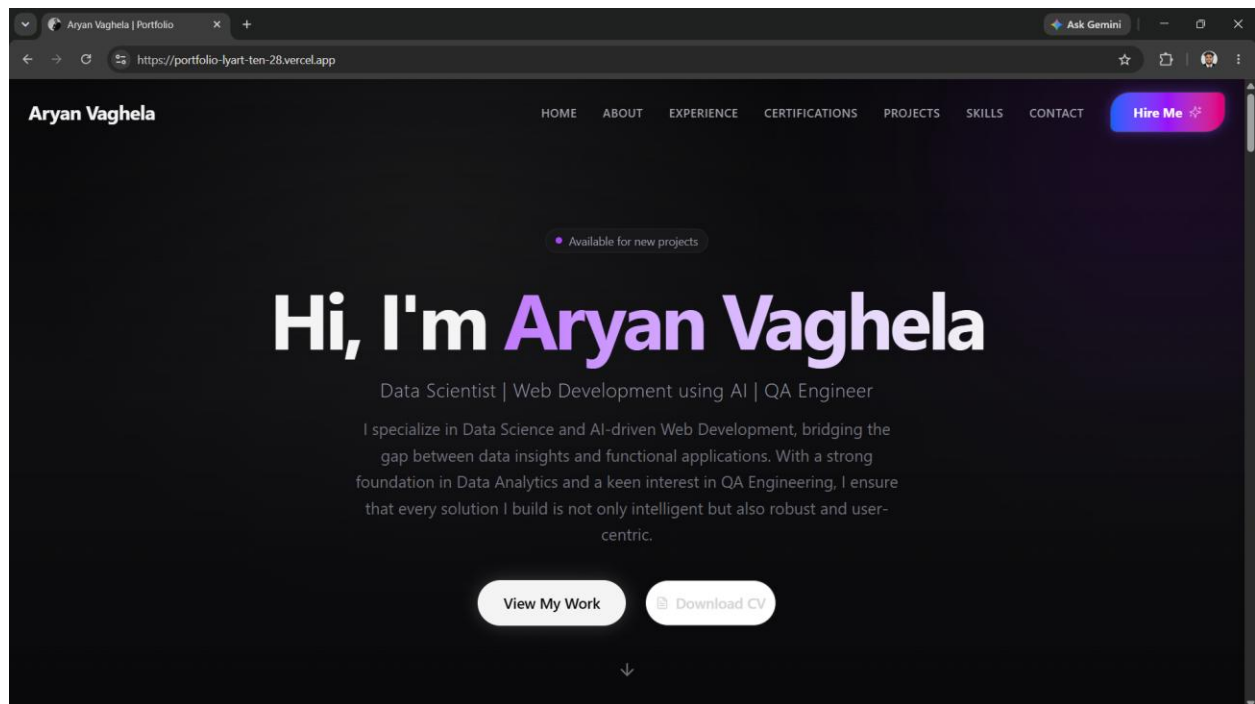


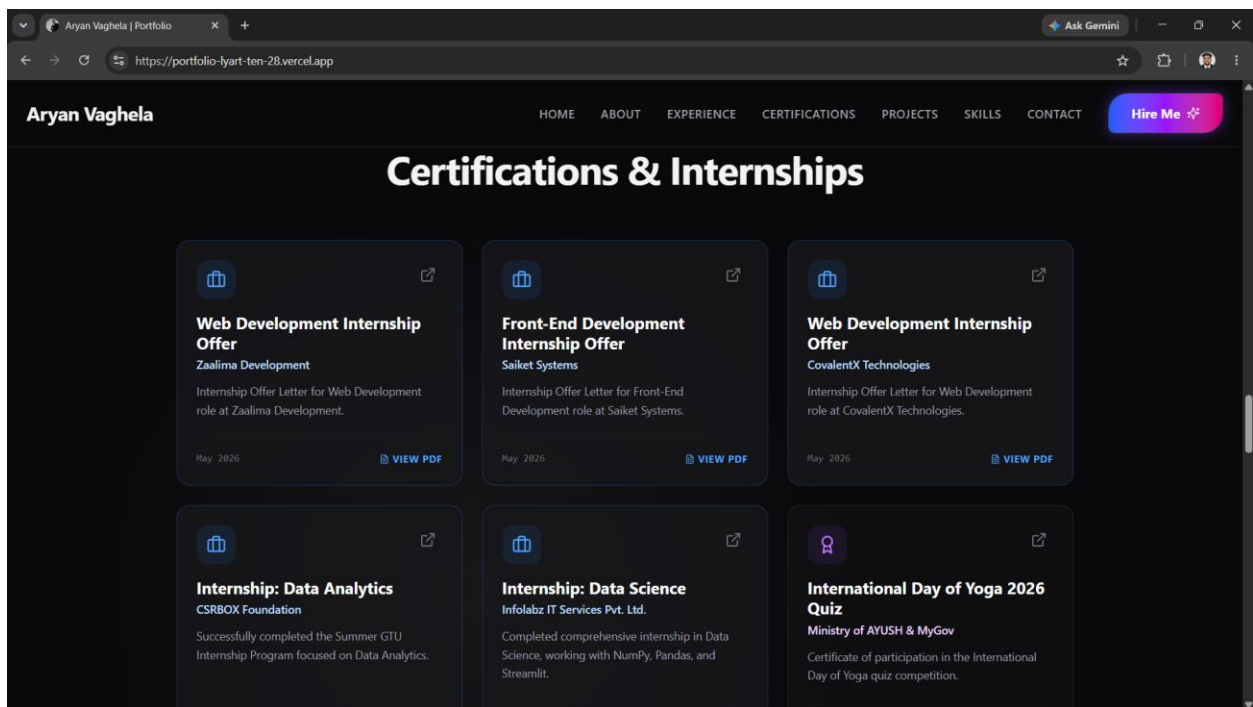
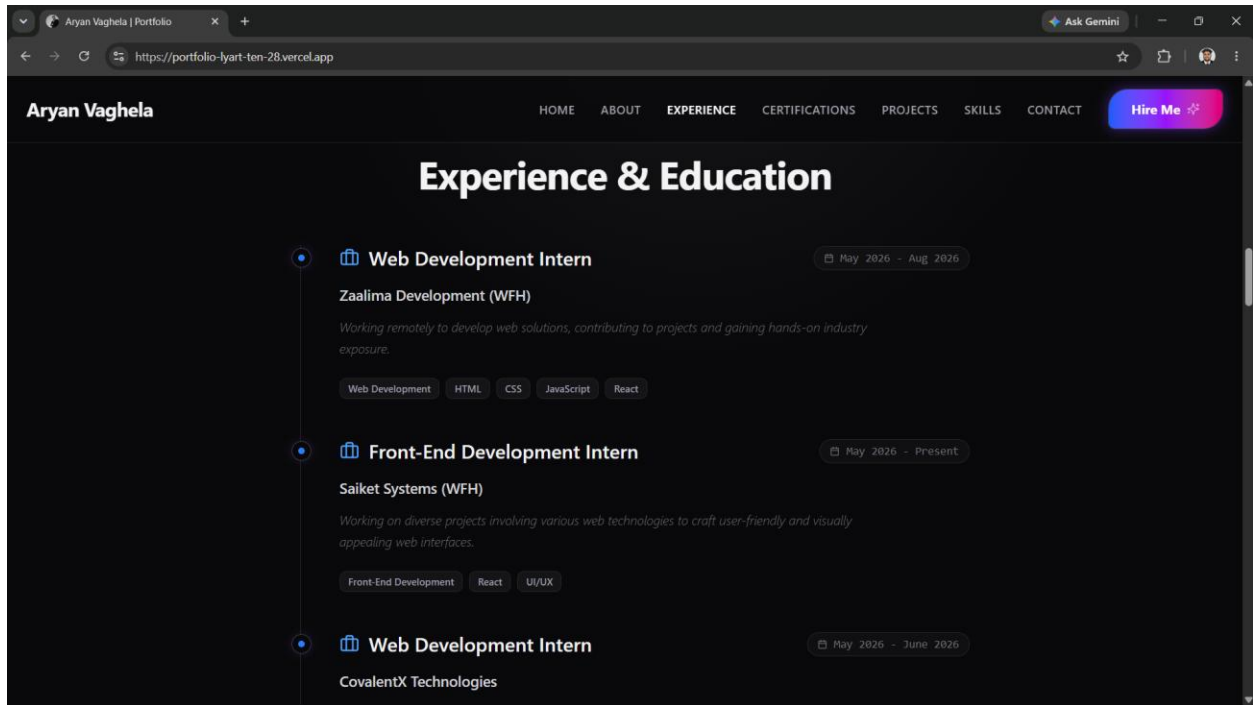


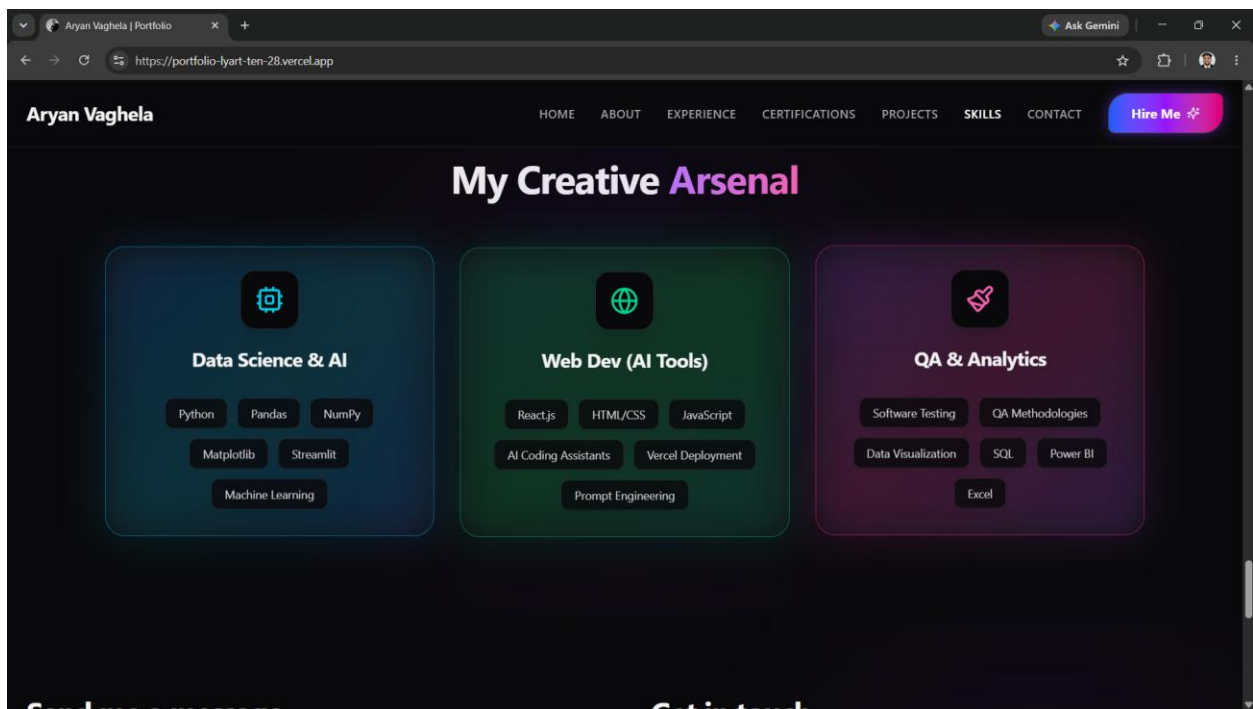
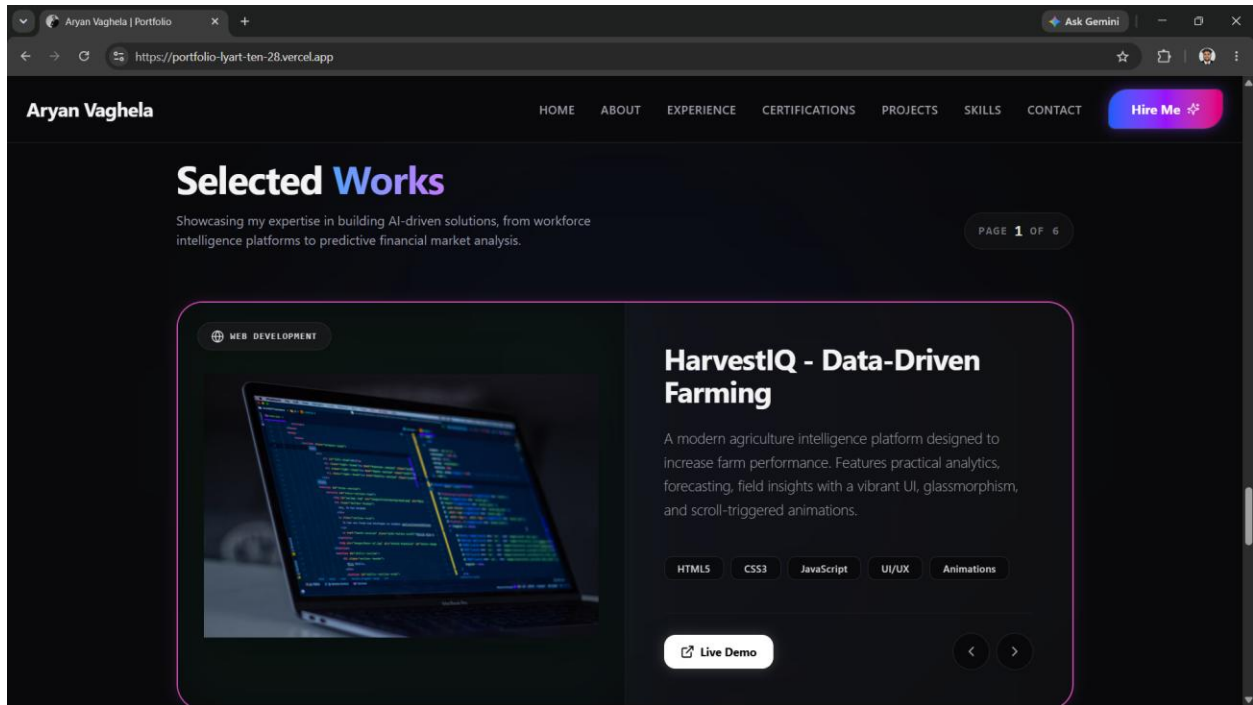
TASK 4:-



TASK 5:-







Aryan Vaghela | Portfolio

Ask Gemini

https://portfolio-lyart-ten-28.vercel.app

Aryan Vaghela

HOMEABOUTEXPERIENCECERTIFICATIONSPROJECTSSKILLSCONTACT

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